Endpoint Security Monitoring using Sysmon on Windows

This project demonstrates the deployment and configuration of Sysmon to enhance endpoint visibility, detect suspicious activity, and generate forensic-level logs for security analysis.

Project Overview

The objective of this project is to implement a system-level monitoring solution capable of capturing detailed telemetry on process creation, network connections, and file system activity. Sysmon was installed, configured, and validated on a Windows environment to simulate a real-world security monitoring setup.

Project Objective

The goal of this project is to:

- Install and configure Sysmon for advanced system monitoring
- Enable detailed logging of process, network, and file activity
- Apply a trusted configuration to improve detection capabilities
- Demonstrate practical endpoint monitoring skills relevant to SOC environments

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Environment and Tools

- Windows Operating System
- PowerShell
- Sysmon
- Windows Event Viewer
- Sysmon community configuration (SwiftOnSecurity)

Project Implementation

1. Navigating to the Working Directory

The Sysmon package was located in the Downloads folder and accessed via PowerShell:

```
cd ~\Downloads\Sysmon
```

---

2. Verifying Sysmon Files

The directory contents were confirmed using:

ls

Files present included:

- Sysmon.exe
- Sysmon64.exe
- Eula.txt
- sysmonconfig-export.xml

```
PS C:\Users\HP\Downloads\Sysmon> ls


    Directory: C:\Users\HP\Downloads\Sysmon


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        3/30/2026   1:12 AM           7490 Eula.txt
-a----        3/30/2026   1:12 AM        6260160 Sysmon.exe
-a----        3/30/2026   1:12 AM        3248064 Sysmon64.exe
-a----        3/30/2026   1:12 AM        3138496 Sysmon64a.exe
-a----        3/30/2026   1:57 AM         123257 sysmonconfig-export.xml
```

3. Downloading the Configuration File

A security-hardened configuration file was downloaded using PowerShell:

Invoke-WebRequest -Uri "https://raw.githubusercontent.com/SwiftOnSecurity/sysmon-config/master/sysmonconfig-export.xml" -OutFile "sysmonconfig-export.xml"

This configuration enhances logging by defining rules for:

- Process creation
- Network connections
- File system activity
- Registry modifications

4. Installing Sysmon

Sysmon was installed using administrative privileges:

.\Sysmon64.exe -i

Output confirmed:

- Sysmon driver installation
- Service startup
- Successful initialization

```
PS C:\Users\HP\Downloads\Sysmon> .\Sysmon64.exe -i


System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
```

5. Applying the Configuration

After installation, the configuration was applied:

.\Sysmon64.exe -c sysmonconfig-export.xml

The system validated the configuration:

- Schema version matched
- Configuration successfully loaded

```
PS C:\Users\HP\Downloads\Sysmon> .\Sysmon64.exe -c sysmonconfig-export.xml


System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.91
Configuration file validated.
Configuration updated.
```

6. Verification and Monitoring

Checking Sysmon Service Status

Get-Service Sysmon64

Output:

- Status: Running

```
PS C:\Users\HP\Downloads\Sysmon> Get-Service Sysmon64

Status   Name              DisplayName
------   ----              -----------
Running  Sysmon64          Sysmon64
```

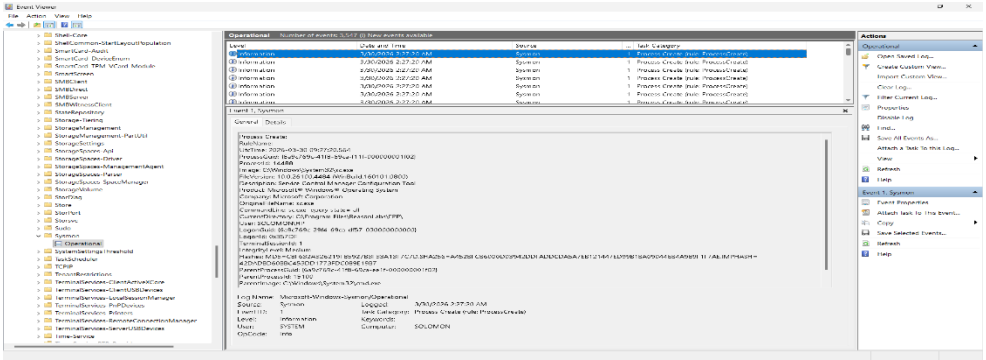
7. **Viewing Logs in Event Viewer**

Logs were accessed via:

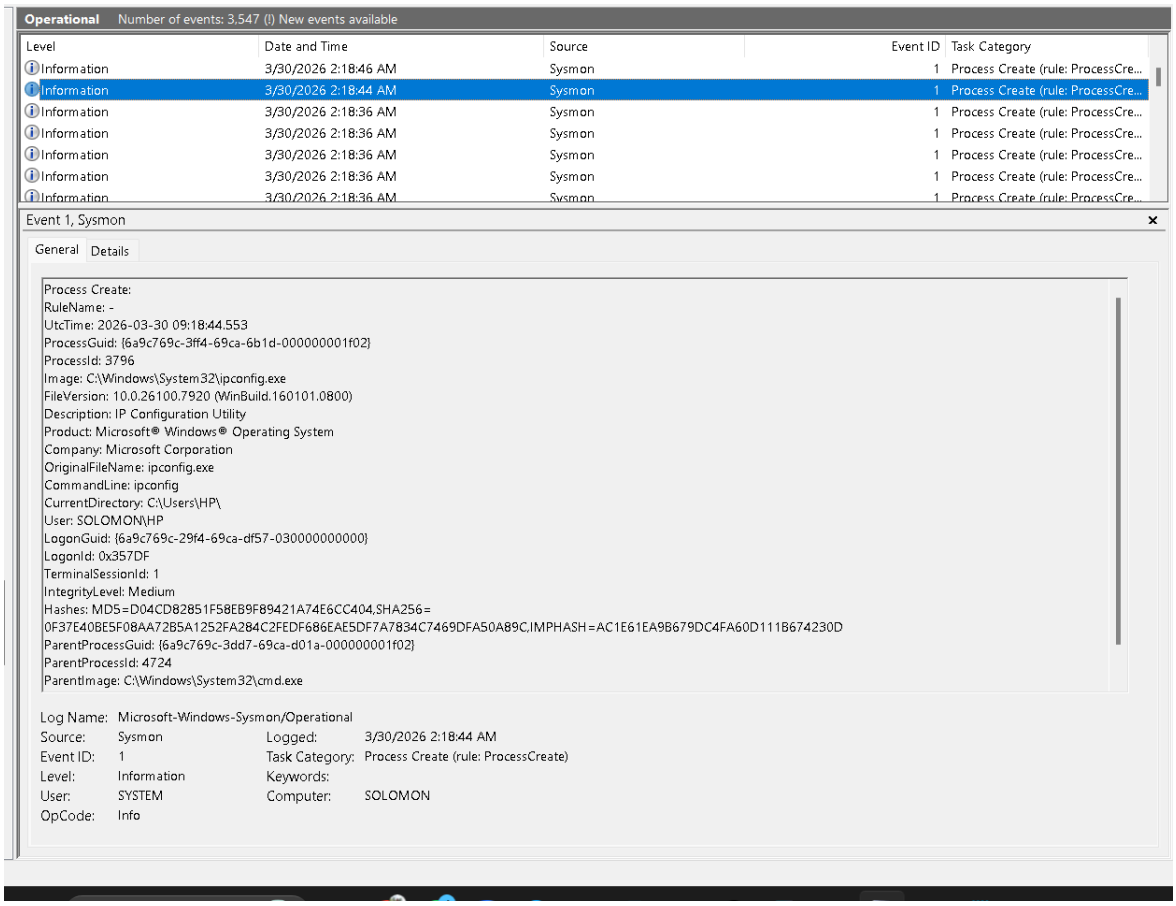
Event Viewer
→ Applications and Services Logs
→ Microsoft
→ Windows
→ Sysmon
→ Operational

Observed event types include:

- Process creation events
- Network connection events
- File creation events



1. Example Sysmon Log Analysis



Operational Number of events: 3,547 (!) New events available

Level	Date and Time	Source	Event ID	Task Category
Information	3/30/2026 2:18:46 AM	Sysmon	1	Process Create (rule: ProcessCre...
Information	3/30/2026 2:18:44 AM	Sysmon	1	Process Create (rule: ProcessCre...
Information	3/30/2026 2:18:36 AM	Sysmon	1	Process Create (rule: ProcessCre...
Information	3/30/2026 2:18:36 AM	Sysmon	1	Process Create (rule: ProcessCre...
Information	3/30/2026 2:18:36 AM	Sysmon	1	Process Create (rule: ProcessCre...
Information	3/30/2026 2:18:36 AM	Sysmon	1	Process Create (rule: ProcessCre...
Information	3/30/2026 2:18:36 AM	Sysmon	1	Process Create (rule: ProcessCre...

Event 1, Sysmon

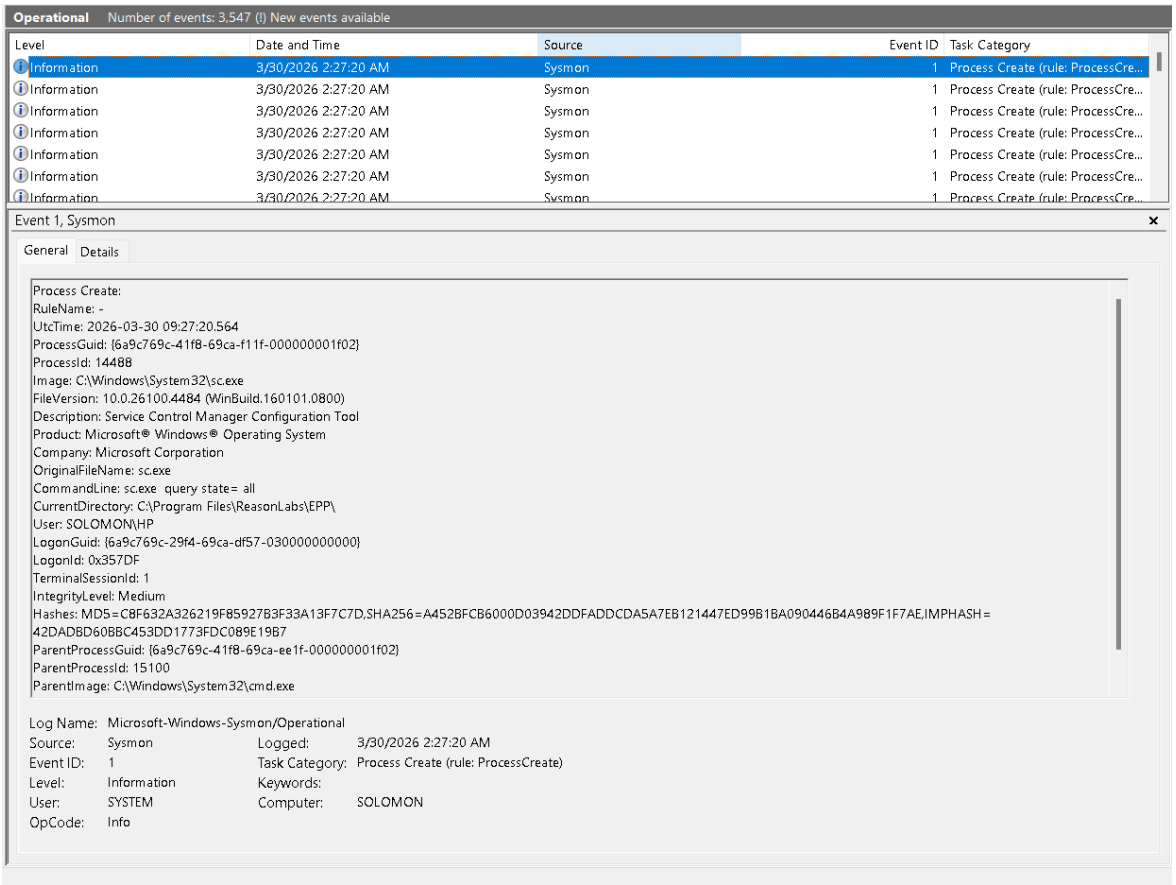
General Details

Process Create:
RuleName: -
UtcTime: 2026-03-30 09:18:44.553
ProcessGuid: {6a9c769c-3ff4-69ca-6b1d-000000001f02}
ProcessId: 3796
Image: C:\Windows\System32\ipconfig.exe
FileVersion: 10.0.26100.7920 (WinBuild.160101.0800)
Description: IP Configuration Utility
Product: Microsoft® Windows® Operating System
Company: Microsoft Corporation
OriginalFileName: ipconfig.exe
CommandLine: ipconfig
CurrentDirectory: C:\Users\HP\
User: SOLOMON\HP
LogonGuid: {6a9c769c-29f4-69ca-df57-030000000000}
LogonId: 0x357DF
TerminalSessionId: 1
IntegrityLevel: Medium
Hashes: MD5=D04CD82851F58EB9F89421A74E6CC404,SHA256=
0F37E40BE5F08AA72B5A1252FA284C2FEDF686EAE5DF7A7834C7469DFA50A89C,IMPHASH=AC1E61EA9B679DC4FA60D111B674230D
ParentProcessGuid: {6a9c769c-3dd7-69ca-d01a-000000001f02}
ParentProcessId: 4724
ParentImage: C:\Windows\System32\cmd.exe

Log Name:	Microsoft-Windows-Sysmon/Operational		
Source:	Sysmon	Logged:	3/30/2026 2:18:44 AM
Event ID:	1	Task Category:	Process Create (rule: ProcessCreate)
Level:	Information	Keywords:	
User:	SYSTEM	Computer:	SOLOMON
OpCode:	Info		

This event demonstrates the successful capture of a process execution event by Sysmon. The ipconfig.exe process was initiated via the Command Prompt by the user SOLOMON\HP. The parent-child relationship confirms direct execution from the command line, illustrating Sysmon's ability to track process lineage and system activity.

2. Example Sysmon Log Analysis



Operational Number of events: 3,547 (!) New events available

Level	Date and Time	Source	Event ID	Task Category
Information	3/30/2026 2:27:20 AM	Sysmon	1	Process Create (rule: ProcessCre...
Information	3/30/2026 2:27:20 AM	Sysmon	1	Process Create (rule: ProcessCre...
Information	3/30/2026 2:27:20 AM	Sysmon	1	Process Create (rule: ProcessCre...
Information	3/30/2026 2:27:20 AM	Sysmon	1	Process Create (rule: ProcessCre...
Information	3/30/2026 2:27:20 AM	Sysmon	1	Process Create (rule: ProcessCre...
Information	3/30/2026 2:27:20 AM	Sysmon	1	Process Create (rule: ProcessCre...
Information	3/30/2026 2:27:20 AM	Sysmon	1	Process Create (rule: ProcessCre...

Event 1, Sysmon

General Details

```
Process Create:
RuleName: -
UtcTime: 2026-03-30 09:27:20.564
ProcessGuid: {6a9c769c-41f8-69ca-f11f-000000001f02}
ProcessId: 14488
Image: C:\Windows\System32\sc.exe
FileVersion: 10.0.26100.4484 (WinBuild.160101.0800)
Description: Service Control Manager Configuration Tool
Product: Microsoft® Windows® Operating System
Company: Microsoft Corporation
OriginalFileName: sc.exe
CommandLine: sc.exe  query state= all
CurrentDirectory: C:\Program Files\ReasonLabs\EPP\
User: SOLOMON\HP
LogonGuid: {6a9c769c-29f4-69ca-df57-030000000000}
LogonId: 0x357DF
TerminalSessionId: 1
IntegrityLevel: Medium
Hashes: MD5=C8F632A326219F85927B3F33A13F7C7D,SHA256=A452BFCB6000D03942DDFADDCDA5A7EB121447ED99B1BA090446B4A989F1F7AE,IMPHASH=
42DADBD60BBC453DD1773FDC089E19B7
ParentProcessGuid: {6a9c769c-41f8-69ca-ee1f-000000001f02}
ParentProcessId: 15100
ParentImage: C:\Windows\System32\cmd.exe
```

Log Name:	Microsoft-Windows-Sysmon/Operational		
Source:	Sysmon	Logged:	3/30/2026 2:27:20 AM
Event ID:	1	Task Category:	Process Create (rule: ProcessCreate)
Level:	Information	Keywords:	
User:	SYSTEM	Computer:	SOLOMON
OpCode:	Info		

The Sysmon Event ID 1 log captured the execution of sc.exe query state= all, initiated from the Command Prompt. The process was run under a standard user context and represents a system enumeration activity. While legitimate in administrative contexts, this behavior is also commonly associated with reconnaissance techniques used by attackers to identify active services on a system.

Key Observations

- Sysmon operates at a deep system level, capturing detailed telemetry
- Proper configuration significantly improves detection capabilities
- Logs provide visibility into system behavior critical for threat analysis
- The tool integrates seamlessly with Windows Event Logging

Challenges and Resolutions

Challenge	Resolution
Incorrect command syntax	Corrected PowerShell syntax using .\Sysmon64.exe
Missing configuration file	Downloaded using Invoke-WebRequest
Access issues	Ensured PowerShell was run as Administrator
File path issues	Navigated to correct directory using cd

Security Relevance

This project demonstrates practical knowledge in:

- Endpoint detection and monitoring
- Log analysis and system auditing
- Threat visibility and forensic readiness
- SOC-level monitoring practices

Sysmon is widely used in enterprise environments for detecting:

- Malware execution
- Lateral movement
- Suspicious network activity
- Persistence mechanisms

Conclusion

The successful installation and configuration of Sysmon demonstrates the ability to deploy a professional-grade monitoring tool capable of capturing high-fidelity system events. The integration of a structured configuration file enhances detection capabilities and provides a strong foundation for cybersecurity analysis.